IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – [NUTRIBOT-AI POWERED NUTRITION AGENT]

Domain of Project – [HEALTH CARE/HEALTH AWARNESS]

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
ASRITHA MUTYALA		asrithamutyala789@gmail.com	9959309689

Problem Statement -

Problem Statement : Nutrition Agent

In an era where health awareness is growing, individuals increasingly seek personalized nutrition guidance. However, most existing tools provide generic diet plans, lack real-time adaptability, and fail to consider a person's holistic lifestyle, cultural preferences, allergies, and evolving health conditions. Dieticians and nutritionists also face limitations in scaling personalized consultations due to time and resource constraints.

Generative AI presents an opportunity to solve this through an intelligent, interactive, and adaptive virtual nutrition assistant that can generate dynamic meal plans, recommend smart food swaps, and explain nutritional choices — all tailored to the individual

Proposed Solution -

Proposed Solution: AI-Powered Nutrition Agent

The proposed solution, **NutriBot**, is an AI-powered Nutrition Agent built using **IBM watsonx.ai, IBM Granite models, and IBM BOB**, deployed on IBM Cloud, to provide personalized, real-time dietary guidance.
NutriBot is built as a **Python Flask web application**. **IBM BOB** was used to auto-generate the complete backend, frontend, and configuration code from a single natural-language prompt, connected securely to IBM watsonx.ai
Users chat naturally with the agent and receive personalized nutrition plans based on age, height, weight, gender, fitness goal, and cuisine preference, with a strong focus on Indian food preferences.

Key features:

- **Conversational Chat Interface** – Ask nutrition questions, get instant personalized responses
- **Personalized Meal Planning** – One-day/weekly diet plans tailored to goals (e.g., weight loss)
- **BMI Calculator** – Calculates BMI and factors it into recommendations
- **Nutrition Dashboard & Family Profiles** – Visual insights; manage plans for multiple family members
- **AGENT_INSTRUCTIONS** – A customizable prompt block that tunes the agent's tone, specialization, and safety rules

Technology Used

- **Langflow platform** -For designing and orchestrating multi-agent workflows visually

- **IBM Grainte model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization

- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.

- **RAG–** RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.

- **IBM Watsonx.ai -** For model deployment, embeddings, and AI governance

- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

Langflow component Used -

1) Chat Input – Interface where users enter their queries, preferences, or meal details (text/voice/image).

2) IBM watsonx AI Agent - Processes user input using Granite models to generate personalized nutrition insights and recommendations.

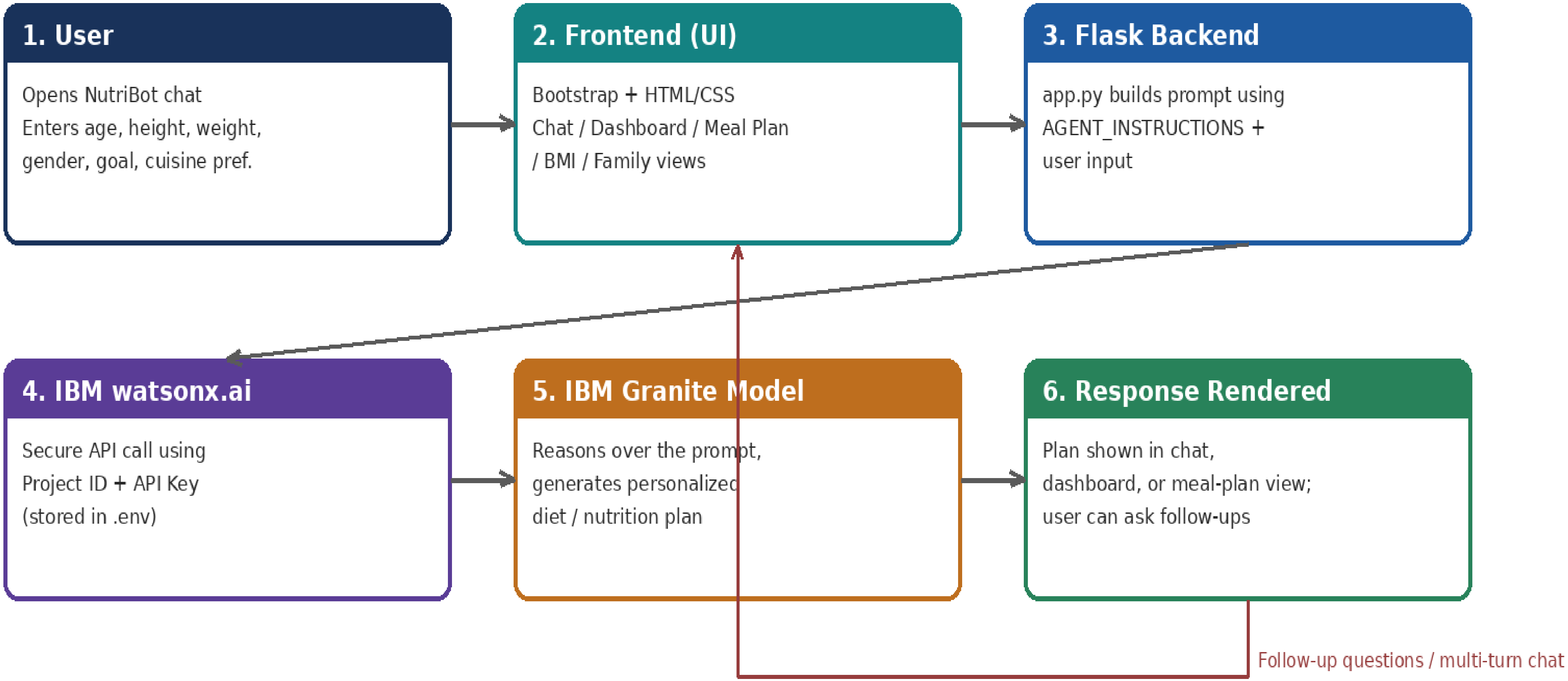
3) Name of IBM Granite model - Granite-4.0-8B-Instruct → Fast, efficient for real-time diet recommendations

4) Chat output -Displays AI-generated responses such as diet plans, nutritional analysis, and health suggestions.

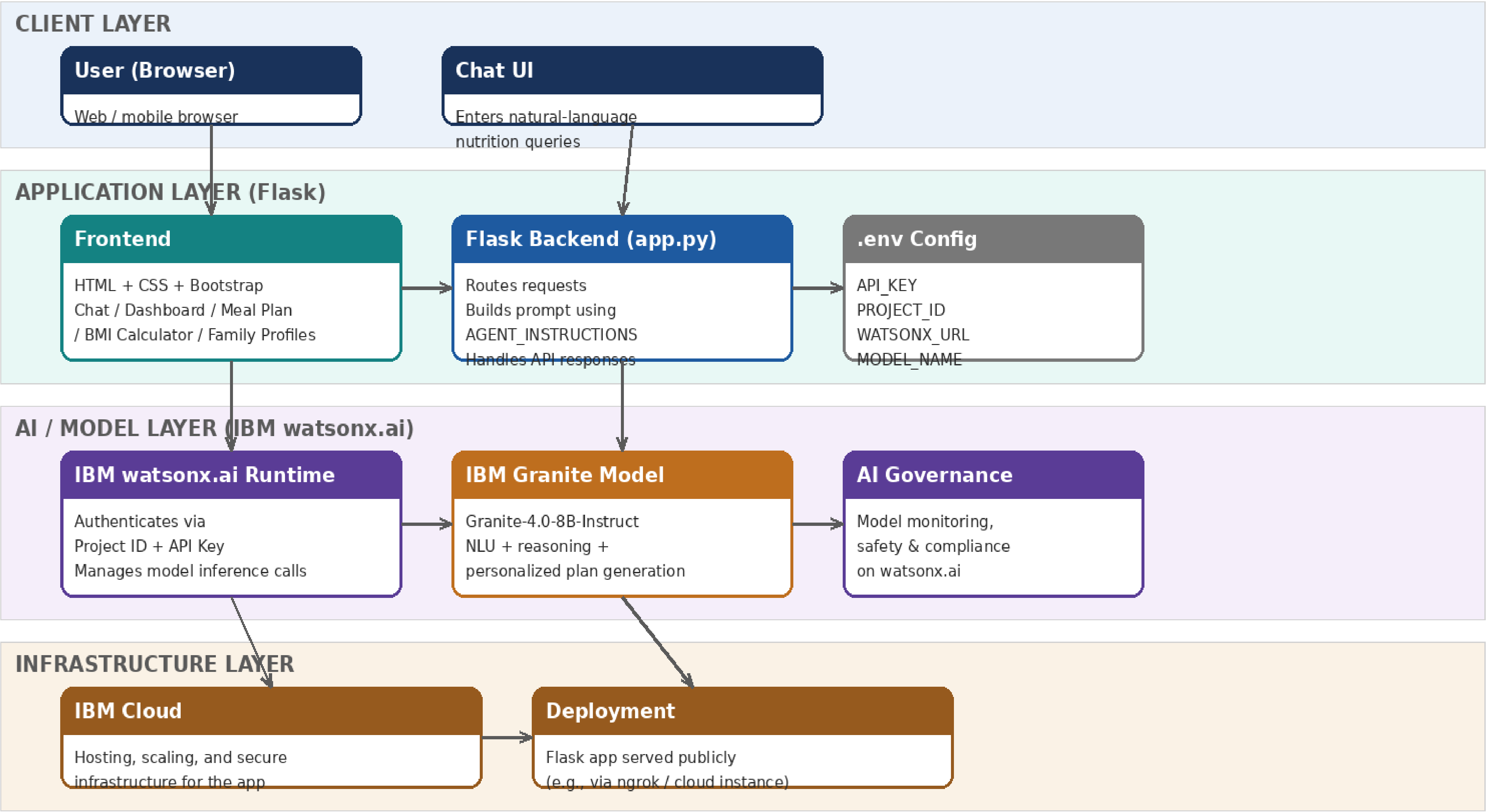
5) .env Configuration File – Securely stores API Key, Project ID, URL, and Model Name

6) AGENT_INSTRUCTIONS Module – Prompt-engineering block that defines the agent's tone, specialization, and safety rules

Langflow workflow -



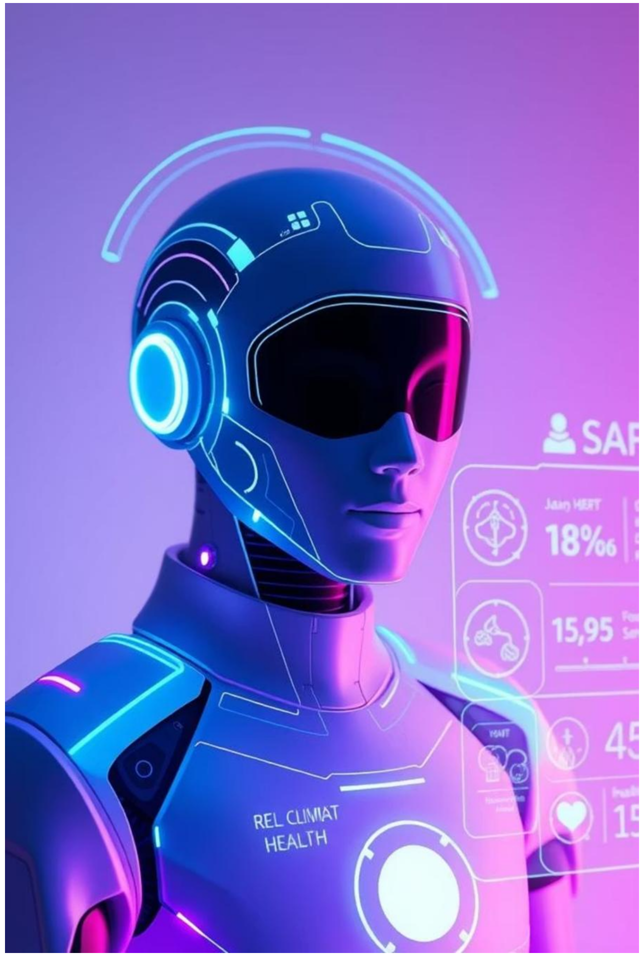
Architecture blueprint



Role of Agentic AI in the solution

- Agentic AI lets NutriBot act as an autonomous nutrition assistant rather than a static form. Built on IBM watsonx.ai and Granite, it interprets natural-language input and reasons over it to generate a complete personalized plan in one response.

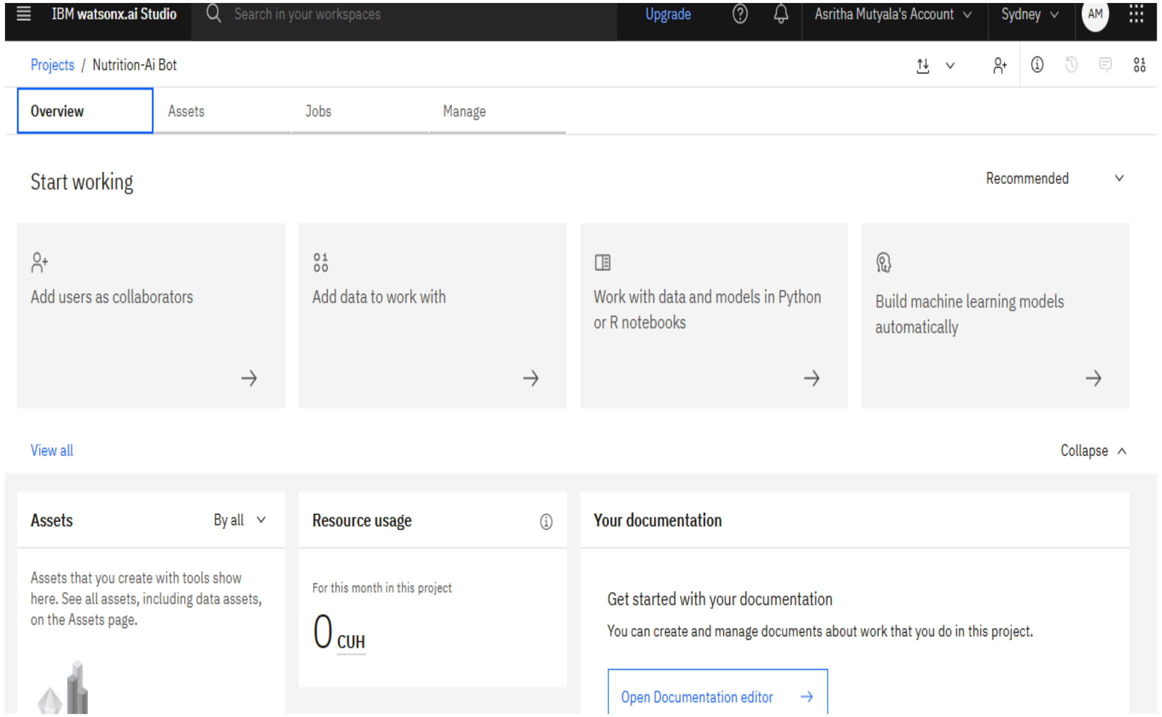
- The AGENT_INSTRUCTIONS block gives the agent a defined persona, specialization, and safety rules, so it consistently applies nutrition guidance without rule-by-rule coding. It also supports multi-turn conversation — users can ask follow-ups ("swap lunch for something vegetarian") and get context-aware answers across chat, dashboard, meal-plan, BMI, and family views.



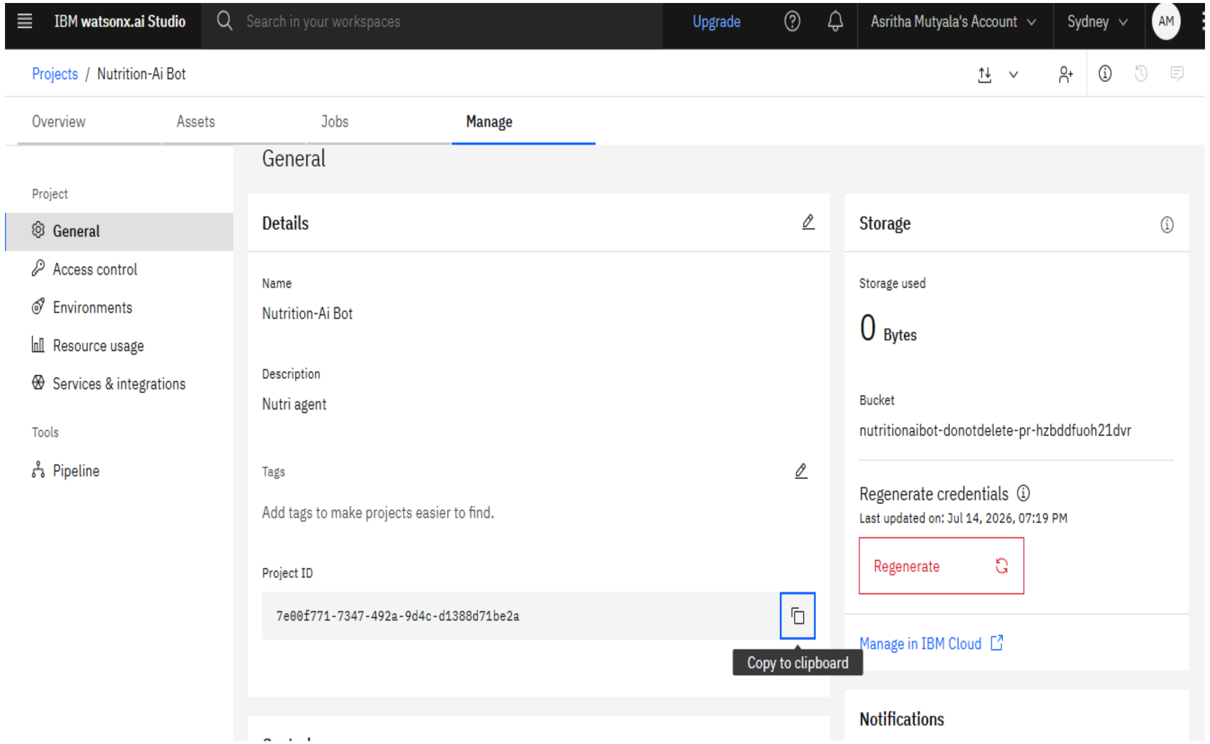
Token Usage

11,761 token are used

Before Prompt



After the Prompt



Novelty and Uniqueness

- **Prompt-to-App Generation** – The entire app was generated from one natural-language prompt via IBM BOB
- **Hyper-Personalization** – Plans factor in age, height, weight, gender, goal, and cuisine (Indian food focus)
- **Customizable Agent Persona** – AGENT_INSTRUCTIONS lets anyone retune tone/specialization without touching core logic
- **Multi-Feature Experience** – Chat + dashboard + meal planning + BMI + family profiles in one app
- **Secure by Design** – API keys kept out of source code via .env
- **Enterprise-Grade Foundation** – Runs on IBM watsonx.ai + Granite on IBM Cloud

Git Hub Link

https://github.com/Ashuu012/Nutribot

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Future Scope

1.Integration with Wearables & Health Devices –

The system can be integrated with smartwatches and fitness trackers to collect real-time data such as steps, calories burned, heart rate, and sleep patterns. This will enable the Nutrition Agent to provide more accurate and dynamic diet recommendations based on actual user activity and health metrics.

2.AI-Powered Voice Assistant & Multilingual Support –

Enhancing the system with a voice-enabled assistant and support for multiple languages will improve accessibility. Users can interact naturally in their preferred language, making the solution more inclusive and suitable for diverse populations, especially in regions with varied linguistic backgrounds.

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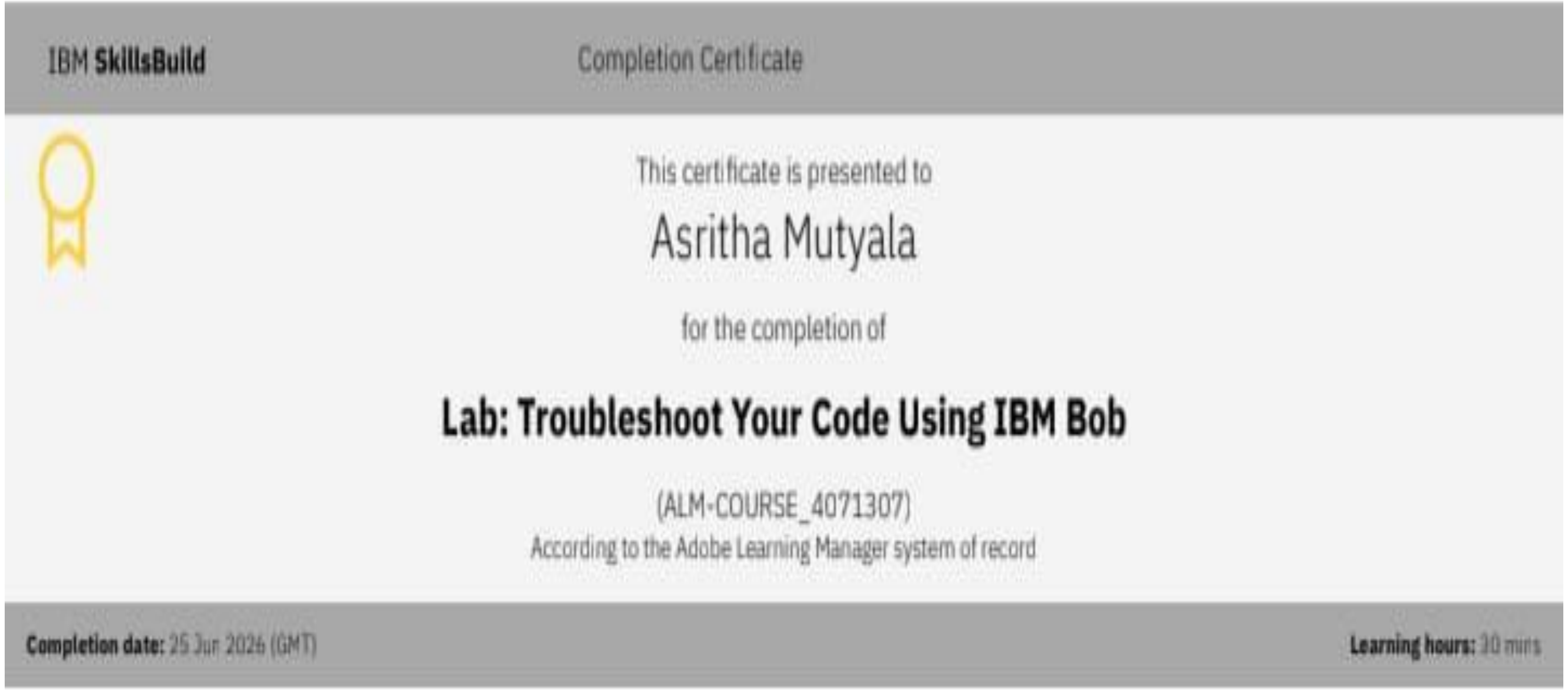

Thank You!

Thank you for your time and interest.